XIANG LI

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EDUCATION

Peking University, Beijing China *Sep. 2018 - Jul. 2023*
Ph.D. in Statistics, School of Mathematical Sciences
Advisor: Prof. Zhihua Zhang

Peking University, Beijing China *Sep. 2014 - Jul. 2018*
B.S. in Statistics, School of Mathematical Sciences
B.A. in Economics, National School of Development

RESEARCH INTERESTS

My research lies at the intersection of statistical machine learning, optimization, and trustworthy AI, with a focus on both theoretical foundations and practical impact. I am particularly interested in developing principled and scalable methods for modern AI systems such as large language models (LLMs). My current research spans the following directions:

- **Statistical Foundations of Generative AI**: Study of the statistical and algorithmic principles underlying generative models. Current work focuses on model evaluation, uncertainty quantification, and model collapse, with the goal of building theoretically grounded tools to better understand and improve model performance.
- **Safety in Generative Models**: Exploration of AI safety challenges in generative models, with an emphasis on language watermarking, robustness to human edits, and detection efficiency. This line of work aims to develop principled mechanisms for tracing model outputs and enabling responsible deployment.
- **Robustness and Privacy in Learning Systems**: Prior work has focused on designing algorithms robust to heavy-tailed noise and adversarial corruption, while ensuring privacy guarantees in both centralized and decentralized settings. Emphasis was placed on approaches that are both provably sound and practically effective.
- **Uncertainty Quantification in Online Learning**: Previous research includes the development of efficient algorithms for streaming and online settings, with rigorous uncertainty quantification, such as confidence intervals for reinforcement learning, federated learning, and adaptive decision-making.
- **Optimization for Decentralized and Federated Learning**: Earlier work studied optimization problems in large-scale distributed systems, with a focus on communication-efficient algorithms and their convergence behavior in federated and decentralized environments.

Optimization: Develop provably optimal methods for (non-)convex and stochastic optimization.

Statistics: Propose practical methods for structural estimation and online statistical inference, and deepen theoretical understanding of the interaction between optimization and sampling.

Sequential Decision Making: Develop computationally efficient online algorithms that balance the exploration-exploitation trade-off in problems such as bandit or reinforcement learning.

Trustworthy Machine Learning: Understand the trade-offs between communication efficiency, accuracy, privacy, robustness, and fairness, and design efficient algorithms with provable guarantees for large-scale problems. Currently, I focus on how to design efficient watermarks and understand its limits for LLMs.

EXPERIENCE

- Aug. 2023 - Now, Post-doctoral researcher at University of Pennsylvania, working with Prof. Weijie Su and Prof. Qi Long.
- Feb. 2022 - Feb. 2023, Visiting Phd. student in Department of Statistical Science at University of Toronto, work with Prof. Qiang Sun.
- 28 Jun. - 10 Jul. 2020, The Machine Learning Summer Schools by the Max Planck Institute for Intelligent Systems, Tübingen, Germany (MLSS 2020, virtual).
- Jun. 2018 - Nov. 2018, Intern at Face ++ (Megvii), Algorithm Group, work with Shuchang Zhou.
- Mar. 2018 - May. 2018, Intern at ToSimple, Algorithm Research Group, work with Naiyan Wang.

PUBLICATIONS

* indicates equal contribution; ** indicates alphabetical order.

Journal papers

[J1] **A Statistical Framework of Watermarks for Large Language Models: Pivot, Detection Efficiency and Optimal Rules** [pdf] [arxiv]
Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, Weijie J. Su
Annals of Statistics, 53 (1): 322-351, Feb. 2025

[J2] **Robust Detection of Watermarks for Large Language Models Under Human Edits [arxiv]**
Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, Weijie J. Su
Major revision at *Journal of the Royal Statistical Society Series B*

[J3] **Optimal Estimation of Watermark Proportions in Hybrid AI-Human Texts** [arxiv]
Xiang Li, Garrett Wen, Weiqing He, Jiayuan Wu, Qi Long, Weijie J. Su
Major revision at *Journal of the American Statistical Association*.

[J4] **Debiasing Watermarks for Large Language Models via Maximal Coupling** [arxiv]
Yangxinyu Xie, **Xiang Li**, Tanwi Mallick, Weijie J. Su, Ruixun Zhang
To appear at *Journal of the American Statistical Association*

[J5] **Convergence and Inference of Stream SGD, with Applications to Queueing Systems and Inventory Control [arxiv]**
Xiang Li*, Jiadong Liang*, Xinyun Chen, Zhihua Zhang
Reject and resubmit at *Operation Research*

[J6] **A Random Projection Approach to Personalized Federated Learning: Enhancing Communication Efficiency, Robustness, and Fairness [pdf]**
Yuze Han**, **Xiang Li****, Shiyun Lin**, Zhihua Zhang**
Journal of Machine Learning Research, 25 (380): 1-88, Dec. 2024

[J7] **Variance-aware Decision Making with Linear Function Approximation with Heavy-tailed Rewards [pdf] [arxiv]**
Xiang Li, Qiang Sun
Transactions on Machine Learning Research, Apr. 2024

[J8] **FedPower: Privacy-Preserving Distributed Eigenspace Estimation [pdf] [arxiv]**
Xiao Guo*, **Xiang Li***, Xiangyu Chang, Shusen Wang, Zhihua Zhang
Machine Learning, 113 (11): 8427-8458, Sep. 2024

Conference papers

[C1] **A Statistical Analysis of Polyak-Ruppert-Averaged Q-Learning** [pdf]
Xiang Li, Wenhao Yang, Jiadong Liang, Zhihua Zhang, Michael I. Jordan
International Conference on Artificial Intelligence and Statistics (AISTATS) 2023

[C2] **Corruption-Robust Variance-aware Algorithms for Generalized Linear Bandits under Heavy-tailed Rewards**
Qingyuan Yu, Euijin Baek, **Xiang Li**, Qiang Sun
Conference on Uncertainty in Artificial Intelligence (UAI) 2025

[C3] **Statistical Analysis of Karcher Means for Random Restricted PSD Matrices** [pdf]
Hengchao Chen, **Xiang Li**, Qiang Sun
International Conference on Artificial Intelligence and Statistics (AISTATS) 2023

[C4] **Statistical Estimation and Online Inference via Local SGD** [pdf]
Xiang Li, Jiadong Liang, Xiangyu Chang, Zhihua Zhang
Conference on Learning Theory (COLT) 2022

[C5] **Asymptotic Behaviors of Projected Stochastic Approximation: A Jump Diffusion Perspective** [pdf]
Jiadong Liang, Yuze Han, **Xiang Li**, Zhihua Zhang
Neural Information Processing Systems (NeurIPS) 2022, Spotlights

[C6] **Personalized Federated Learning towards Communication Efficiency, Robustness and Fairness** [pdf]
Shiyun Lin*, Yuze Han*, **Xiang Li**, Zhihua Zhang
Neural Information Processing Systems (NeurIPS) 2022

[C7] **Communication-Efficient Distributed SVD via Local Power Iterations** [pdf]
Xiang Li, Shusen Wang, Kun Chen, Zhihua Zhang
International Conference on Machine Learning (ICML) 2021

[C8] **Finding the Near Optimal Policy via Reductive Regularization in MDPs** [pdf]
Wenhao Yang, **Xiang Li**, Guangzeng Xie, Zhihua Zhang
Workshop on Reinforcement Learning Theory, ICML 2021

[C9] **On the Convergence of FedAvg on Non-IID Data** [pdf]
Xiang Li*, Kaixuan Huang*, Wenhao Yang*, Shusen Wang, Zhihua Zhang
International Conference on Learning Representations (ICLR) 2020, Oral presentation

[C10] **Do Subsampled Newton Methods Work for High-Dimensional Data?** [pdf]
Xiang Li, Shusen Wang, Zhihua Zhang
AAAI Conference on Artificial Intelligence (AAAI) 2020

[C11] **A Regularized Approach to Sparse Optimal Policy in Reinforcement Learning** [pdf]
Wenhao Yang*, **Xiang Li***, Zhihua Zhang
Neural Information Processing Systems (NeurIPS) 2019

Preprints

[P1] **Evaluating the Unseen Capabilities: How Many Theorems Do LLMs Know?** [arxiv]
Xiang Li, Jiayi Xin, Qi Long, Weijie J. Su
Under review

[P2] **Uncertainty Quantification of Data Shapley via Statistical Inference** [pdf]
Mengmeng Wu, Zhihong Liu, **Xiang Li**, Ruoxi Jia, Xiangyu Chang
Under review

[P3] **Finite-Time Decoupled Convergence in Nonlinear Two-Time-Scale Stochastic Approximation** [arxiv]
Yuze Han**, **Xiang Li****, Zhihua Zhang**
Under review

[P4] **Decoupled Functional Central Limit Theorems for Two-Time-Scale Stochastic Approximation** [arxiv]
Yuze Han, **Xiang Li**, Jiadong Liang, Zhihua Zhang
Under review

[P5] **Online Statistical Inference for Nonlinear Stochastic Approximation with Markovian Data** **[arxiv]**
Xiang Li, Jiadong Liang, Zhihua Zhang
Under review

[P6] **Privacy-Preserving Community Detection for Locally Distributed Multiple Networks** [arxiv]
Xiao Guo, **Xiang Li**, Xiangyu Chang, Shujie Ma
Under review

[P7] **Asymptotic Behaviors and Phase Transitions in Projected Stochastic Approximation: A Jump Diffusion Approach** **[arxiv]**
Jiadong Liang, Yuze Han, **Xiang Li**, Zhihua Zhang
Preprint

[P8] **Communication-Efficient Local Decentralized SGD Methods** **[arxiv]**
Xiang Li, Wenhao Yang, Shusen Wang, Zhihua Zhang
Preprint

PRESENTATIONS

- *Optimal Robust Detection for Gumbel-Max Watermarks Under Contamination*
  - Jul. 2024, IMS-China International Conference on Statistics and Probability, Yinchuan.
  - Sep. 2024, Penn conference on Big Data in Biomedical & Population Health Sciences.
  - Sep. 2024, Warren & ASSET Center Research Mixer, UPenn.
- *A Statistical Framework of Watermarks for Large Language Models: Pivot, Detection Efficiency and Optimal Rules*
  - May 2024, the NUS IMS workshop on statistical machine learning for high dimensional data, Singapore.
  - Jul. 2024, the 2nd joint conference on statistics and data science in China (JCSDS), Kunming.
  - Nov. 2024, the conference on statistical learning and data science (SLDS), Newport Beach, California.
- *Complete Asymptotic Analysis for Projected Stochastic Approximation and Debiased Variants*
  - Sep. 2023, the Allerton conference at the University of Illinois, Allerton Park & Retreat Center.
- *Polyak-Ruppert-Averaged Q-Learning is Statistically Efficient*
  - Jul. 2022, the RL workshop in Shanghai University of Finance and Economics.
- *Statistical Estimation and Inference via Local SGD in Federated Learning*

  - Nov. 2021, the 14-th China-R conference.
- *A Regularized Approach to Sparse Optimal Policy in Reinforcement Learning*
  - Jun. 2019, the machine learning workshop at Peking University.

ACADEMIC SERVICE

- Conference review: NeurIPS, ICML, ICLR, UAI, AISTATS, IJCAI.
- Journal review: Journal of Machine Learning Research (JMLR), Transactions on Machine Learning Research (TMLR), Journal of the American Statistical Association (JASA), Annals of Applied Probability (AOAP), Operational Research (OR), IEEE Transactions on Automatic Control, IEEE Open Journal of Signal Processing, IEEE Journal on Selected Areas in Communications, Transactions on Parallel and Distributed Systems, World Wide Web (WWW).
- Event organization:
  - Aug. 2024, Session Chair for "Theoretical and Algorithmic Developments in Federated Learning" at the Modeling and Optimization: Theory and Applications (MOPTA) conference.

TEACHING EXPERIENCES

- *Reinforcement Learning: Theory and Algorithms*, Fall 2019, PKU, Teaching Assistant
- *Linear Algebra*, Spring 2021, PKU, Teaching Assistant

SELECTED AWARDS

- IMS New Researcher Travel Award *2025*
- Outstanding PhD Graduates, Peking University *2023*
- President Scholarship for Excellent Ph.D Student, Peking University *2018, 2020-2021*
- Outstanding reviewer in NeurIPS, top 8% *2021*
- National Scholarship, China *2017, 2019*
- Travel Award *NeurIPS 2019, AAAI 2020*
- Outstanding Graduates, School of Mathematical Sciences, Peking University *2018*
- First prize in National Undergraduate Mathematical Modeling Contest, China *2016*